Yixuan Shao

PhD Student · Pulsar research

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Education

2023–present **PhD, High-energy astrophysics**, *School of Astronomy and Space Science*, Nanjing University, Advisor: Prof. Ping Zhou.
Working on multiwavelength observations of isolated neutron stars (e.g. magnetars, central compact objects, high-magnetic-field pulsars, radio pulsars), fast radio bursts, and supernova remnants.

2019–2023 : **Bachelor of Astronomy**, *School of Astronomy and Space Science*, Nanjing Univeristy, Undergrad research advisor: Prof. Ping Zhou.
Study pulsar search with FAST & optical/NIR and X-ray data analysis

Publications

See full list on Ads library

2025 **Shao, Yi-Xuan**, Ping Zhou, Xiao Zhang, Zhi-Yu Zhang, Yang Chen, Qin Han, Di Li, Xiang-Dong Li, Jian-Bin Weng, and Yong Shao. Absence of Radio Emission Reveals an Exceptionally Weak Explosion of the Putative Historical Supernova Pa 30. *ApJ*, volume 992, page L6, October 2025.

2024 **Shao, Yi-Xuan**, Ping Zhou, Xiang-Dong Li, Bin-Bin Zhang, Alberto Javier Castro-Tirado, Pei Wang, Di Li, Zeng-Hua Zhang, Zi-Jian Zhang, You-Dong Hu, and Shashi B. Pandey. GTC Optical/Near-infrared Upper Limits and NICER X-Ray Analysis of SGR J1935+2154 for the Outburst in 2022. *ApJ*, volume 976, page 99, November 2024.

2026 Shi-Jie Gao, **Shao, Yi-Xuan**, and Xiang-Dong Li. Pulsar Gleaners: Discovery of 19 Pulsars in FAST Archival Data at $|b| < 5^\circ$ and Decl.$< -5^\circ$. *arXiv e-prints*, page arXiv:2601.01912, January 2026.

2025 Shi-Jie Gao, Xiang-Dong Li, Zhen Yan, **Shao, Yi-Xuan**, and Ping Zhou. RRAT J2325–0530: A Rotating Radio Transient with an Atypical Waiting-time Distribution. *ApJ*, volume 991, page 201, October 2025.

2025 Shi-Jie Gao, Xiang-Dong Li, Song Wang, Kareem El-Badry, De-Jiang Zhou, **Shao, Yi-Xuan**, Zhen Yan, Pei Wang, Ping Zhou, and Jin-Lin Han. Search for Radio Pulsations from Neutron Star Candidates in Detached Binaries. *arXiv e-prints*, page arXiv:2511.20242, November 2025.

2025 Shi-Jie Gao, Xiang-Dong Li, **Shao, Yi-Xuan**, Ping Zhou, Pei Wang, Yun-Wei Yu, Zhen Yan, and Di Li. A Target Search for Fast Radio Bursts Associated with Two Fast Blue Optical Transients: AT2018cow and CSS161010. *ApJ*, volume 990, page 93, September 2025.

2024 Yi-Han Iris Yin, Bin-Bin Zhang, Jun Yang, Hui Sun, Chen Zhang, **Shao, Yi-Xuan**, You-Dong Hu, Zi-Pei Zhu, Dong Xu, Li An, He Gao, Xue-Feng Wu, Bing Zhang, Alberto Javier Castro-Tirado, Shashi B. Pandey, Arne Rau, Weihua Lei, Wei Xie, Giancarlo Ghirlanda, and et al. Piro. Triggering the Untriggered: The First Einstein Probe-detected Gamma-Ray Burst 240219A and Its Implications. *ApJ*, volume 975, page L27, November 2024.

2024 Wan-Jin Lu, Ping Zhou, Pei Wang, **Shao, Yi-Xuan**, Xiang-dong Li, Jacco Vink, Di Li, and Yang Chen. Upper Limits on the Radio Pulses from Magnetars and a Central Compact Object with FAST. *ApJ*, volume 963, page 151, March 2024.

2024 Shi-Jie Gao, **Shao, Yi-Xuan**, Pei Wang, Ping Zhou, Xiang-Dong Li, Lei Zhang, Joseph W. Kania, Duncan R. Lorimer, and Di Li. Discovery of a Millisecond Pulsar Associated with Terzan 6. *ApJ*, volume 974, page L2, October 2024.

Presentations

2024.12 **A pipeline to search for special magneto-active regions near Fast radio bursts**, FRB 2024, Khao Lak, Thailand

2024.12 **GTC Optical/Near-infrared Upper Limits and NICER X-Ray Analysis of SGR J1935+2154 for the Outburst in 2022**, PSR 2024, Guizhou, China

2025.8 **The absence of radio emission reveals an exceptionally weak explosion for the probable historical supernova Pa 30**, The 16th Zhang Heng Academic Symposium, Qingdao, China

Research Experience

Nanjing Univerisity, *Advisior: Prof. Ping Zhou*

2023.9 – present ***Multiwavelength observation of isolated neutron stars***.
Utilizing NICER (X-ray), GTC/VLT (optical and near-infrared), and FAST (radio) to search for and study radiative emission from isolated neutron stars (focus on magnetars, central compact objects (CCOs), and high-magnetic-field pulsars). (Shao et.al, 2024, ongoing observations and manuscripts in preparation...)

2024.7 – 2025.9 ***VLA observation of a remnant from double-white dwarf merger, Pa30***.
Analyzing the VLA imaging data, search for the extended radio emission, study the true nature of Pa30, SNR or a wind nebula? (Shao et.al, 2025)

Tsinghua Univerisity, *Advisior: Prof. Dongzi Li*

2024.7 – present ***FRB local environment and FAST single pulse polarization study***.
Study the neighbor plasma environment of some FRB source through polarization observation. Study the single pulse polarization characteristics, which may reveal the magnetosphere configuration. (1 manuscript to be submitted and ongoing observations...)

Awards & Scholarships

2024 Xiaomi Scholarship
2023 Outstanding Graduate, Nanjing University
2022 Xiangyun Scholarship
2022 Outstanding Award, Fundamental Sciences Scholarship, Nanjing University
2021 Yang Lanyun Leadership Excellence Scholarship
2020 Zheng Gang Elite Scholarship
2020 Scholarship, National Astronomical Observatories of China

Awarded Telescope time (PI) till 2025.12

FAST 16h; 2023,2024,2025
GTC 1.5h; 2022
VLT 1.5h; 2025
XMM-Newton 6h; 2026

Computer skills

Programming Languages: Python, Bash (daily use); C/C++ and Fortran (occasional use)

Teaching Assistantship

Fall, 2024 : **General Astronomy**.

Advisors & Collaborators

Ping Zhou
Associate Professor
School of Astronomy and Space Science
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Dongzi Li
Assistant Professor
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